Marmara University
Electrical and Electronics Engineering Department
Spring 2025
EE 3011 – Electronics I
Course Project – Due May 31, 2025

Frequency-to-voltage Converter

The main purpose of this project is to study and verify the operation of a frequency-to-voltage converter electronic circuit. A frequency-to-voltage converter is a circuit whose output voltage is proportional to the frequency of a sinusoidal input. This circuit finds many applications in several fields including communications and instrumentation. The study includes simulation verification and hardware implementation.

Specifications

- Search the internet for a reliable and simple frequency-to-voltage converter circuit. Pick a circuit that contains at least one discrete transistor (MOSFET or BJT). Also, avoid circuits that contain complicated (and probably unavailable) parts and cannot be simulated using LTspice.
- Make sure you understand the circuit and the function of each component. Also, in case some components were not available, consider replacing them with alternatives.
- The circuit is supposed to work within a specified frequency range. Based on your study, define the frequency range of interest for your design.

Instructions

- LTspice simulation of the circuit (50%). The simulation should cover different aspects of the circuit function, as follows.
    a) Circuit function analysis. In particular, frequency-voltage relationship.
    b) Circuit limitations, range of operation and best operating conditions.
    c) Possible modifications/improvements.
- Hardware implementation of the circuit (50%). Implement the circuit and make studies on it. In particular, produce a curve showing the relationship between the frequency of the input signal and output voltage.
- Maximum of two (2) students can work as a team.
- Access to the electronics lab can be arranged, in case a signal generator is needed.
- When your circuit is ready, take an appointment for circuit evaluation.
- Submit a report on your simulation study and investigations of your circuit. The report should also include the hardware implementation details. Avoid lengthy introductions and unnecessary details.
- Prepare your project report (all in one file) in pdf format and upload to the site below.

Upload link: https://www.dropbox.com/request/nQlDq3EtYtptvyoilTJN

Show all work and analysis
Only pdf files are acceptable
Late submissions are checked but not graded
Copied submissions represent academic dishonesty and are disqualified